ABDELRAHMAN ELGRM

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EXECUTIVE SUMMARY

Data Scientist and Python developer with hands-on experience building end-to-end ML/DL pipelines, computer vision systems, and NLP solutions using TensorFlow, PyTorch, and Scikit-learn. Designed and deployed 5 production-level projects spanning RAG-based LLM agents, CNN quality control systems, and real-time recommendation engines. Currently pursuing a degree in Electrical and Computer Engineering. Seeking an AI/Data Science internship to solve real-world problems with intelligent, scalable solutions.

EDUCATION

Menofia University Oct. 2022 – Jan. 2027
Bachelor in Electrical and Computer Engineering ***GPA: 3.47***

PROFESSIONAL EXPERIENCE

ELARABY GROUP Jul. 2025 – Sept. 2025
Engineering Intern

- Developed a cross-platform Python/PyQt application for factory automation and engineered a CV model to detect adhesive precision on LED TV backlights.
- Designed loss prevention systems and conducted technical audits to optimize manufacturing workflows.

ALX Africa Jan. 2025 – Jun. 2025
Data Scientist Intern

- Engineered end-to-end ML/DL, Computer Vision, and NLP solutions using TensorFlow and PyTorch alongside automated Python/SQL data pipelines and Power BI dashboards.
- Managed the full data lifecycle—from preprocessing to deployment—translating complex industrial datasets into actionable KPIs for stakeholders.

TECHNICAL PROJECTS

NEURASHIELD AI-SOC Analyst | ***Python, FastAPI, TensorFlow, Supabase, LLMs, RAG, SQL***

- Engineered an automated SOC agent using **FastAPI** and **Supabase** to ingest and normalize raw security logs for real-time threat detection. Integrated a **RAG** pipeline to contextualize anomalies against industry-standard cybersecurity frameworks.
- Developed auto-generated incident response playbooks using **fine-tuned LLMs** and utilized **TensorFlow** with vector search to optimize retrieval of security intelligence.

Feasibility Study Desktop App | ***Python, PyQt6, QThread***

- Engineered a multithreaded desktop application for industrial feasibility studies, utilizing **QThread** to maintain UI responsiveness during heavy data processing.
- Designed an intuitive **PyQt6** interface to streamline data entry and report generation for engineering workflows.

Computer Vision for Manufacturing | ***OpenCV, TensorFlow, CNNs***

- Developed a dual-purpose **CNN** system to automate assembly line quality control, reducing defect escape rate.
- Implemented real-time object tracking using **OpenCV** for security monitoring across manufacturing facilities.

Maji Ndogo Water Crisis Analysis | ***SQL, Python, Power BI***

- Analyzed **60,000+** records via **SQL** to identify critical infrastructure failures across water distribution networks.
- Developed interactive **Power BI** dashboards to visualize resource gaps and optimize distribution strategies for stakeholders.

Movie Recommendation System | ***Python, Scikit-learn, Pandas, Flask***

- Built a hybrid **collaborative and content-based filtering** recommendation engine using **Scikit-learn**, achieving high prediction accuracy on user preference datasets.
- Deployed the model via a **Flask** REST API, enabling real-time personalized movie recommendations through an interactive web interface.

COURSES & CERTIFICATIONS

- **ML Specialization** (DeepLearning.AI): Supervised ML (regression, classification, gradient descent) · Advanced Algorithms (MLPs, ensemble models, TensorFlow) · Unsupervised Learning (K-means, PCA, Deep Q-Networks).
- **CS109: Probability for Computer Scientists** (Stanford): Mastered statistical inference and probabilistic modeling to optimize ML performance.
- **Machine Learning Engineer Track** (DataCamp): End-to-end ML engineering covering model development, deployment, and production pipelines.

SKILLS & LANGUAGES

Technical Skills: Python · SQL · TensorFlow · PyTorch · OpenCV · PyQt6 · FastAPI · Flask · Scikit-learn · Pandas · NumPy · Power BI · Supabase · PostgreSQL · Git · LLMs · RAG · Vector Database

Soft Skills: Agile Collaboration · Quantitative Reasoning · Impactful Communication · Technical Auditing · Critical Problem Solving · Project Management · Adaptability · Strategic Planning · Cross-functional Teamwork

Languages: Arabic (Native) · English (Fluent) · German (Beginner)